Problem 1 ... rotating a force

3 point

There are two forces $F_1 = 2.0\,\mathrm{N}$ and $F_2 = 1.0\,\mathrm{N}$ acting on a point of mass. What is the angl
between them if their resultant is the same magnitude as the larger of the forces, $F = F_1$?

May the Force be with you

Problem 2 ... five-second rule

3 point

You may have heard that if you drop your food on the floor but pick it up within five seconds t will not be heavily contaminated with bacteria. Let us consider the following case. Yo drop a circular snack with a diameter of 4 cm *on the ground. The bacteria from the floo will immediately stick to it. However, according to the common rule, this should not matte Therefore, let us assume that most of the bacteria only arrive at the food from the vicinity o he snack during those five seconds. What would their velocity have to be for their numbers t multiply tenfold on a snack in five seconds? The surface density of bacteria on the ground i homogeneous. Jarda always blows off the fallen food so that he can eat with a peace in mind*

Problem 3 ... game of tag

3 point

Two cars are driving on a road parallel to each other in the same direction. The trajectorie of the cars are $\xi = 1.5\,\mathrm{m}$ apart. Nicolas is $d = 3\,\mathrm{m}$ away from the trajectory of the first car which is moving at $v_1 = 55\,\mathrm{km{\cdot}h^{-1}}$. What is the velocity of the second car if it always stay hidden behind the first car from Nicolas's point of view? We approximate the cars as poin masses.

Nicolas waited for far too long at the bus stop

Problem 4 ... changing the transmission lines

3 point

We have a transmission line with high voltage $U_0 = 110\,\mathrm{kV}$, and we would like to increase it t
xtra-high voltage $U_1 = 400\,\mathrm{kV}$. Assuming the resistance of the line is constant, how much wi
he power losses on the line change? We are interested in the power loss ratio P_1/P_0.

Karel thought about changing the transmission lines

Problem 5 ... swimming problem

3 point

Verča would like to go swimming, but she dislikes going into the water without eating. She ha an average density of $945\,\mathrm{kg{\cdot}m^{-3}}$ *when she is hungry, and it is hard for her to dive. How man kilograms of food does Verča need to eat to have an average density of at least* $980\,\mathrm{kg{\cdot}m^{-3}}$ *Assume that she does not change her volume when she eats. The length of the pool is* 30 m *and its depth is* 3.1 m*. Verča's weight before the meal is* 47 kg*.*

Verča sometimes feels very empty inside

Problem 6 ... slide

3 point

The management of the Dormitories and Refectories decided to spend money meaningfully, s
hey built a slide from the roof of the "building A" of the 17th November dormitories directl
o the door of the MFF's Impakt pavilion. The two buildings are 430 m *apart as the crow flies*
nd their height difference is 59 m*. What is the coefficient of friction of the slide if a studen*
veighing 60 kg *has thrown himself down it with a velocity* $v_0 = 8.5\,\mathrm{m\cdot s^{-1}}$ *and just stops at th*
nd of it? Assume that the slide is an inclined plane. *Eliška was late for a lecture*

Problem 7 ... Jarda's problems

3 point

When selecting problems for Physics Brawl Online, Jirka calculated that 20 % of the availabl problem assignments comes from Jarda. Yet, Jarda is the author of an incredible third of a the problems selected for the competition. Let us assume that FYKOS organizers do read th problem proposals when selecting them for the competition. How much more likely is it tha the organizers will select a problem if the author is Jarda than if the proposal comes fro another (average) organizer?

Jirka's original version of the problem did not pass the political censorship

Problem 8 ... irresistibly attractive

4 point

Jindra can't find a girlfriend, so he orders a female-attracting device from a dubious onlin store. He received a pocket black hole. At a distance of $5\,\mathrm{m}$*, the black hole exerts gravitationa acceleration of* $9.81\,\mathrm{m{\cdot}s^{-2}}$ *on all bodies (including girls). Calculate the Schwarzschild radius o his black hole. Jindra thought about complaining, but the black hole swallowed him*

Problem 9 ... flying droplets

4 point

Rotating around a horizontal axis is a wheel with an outer diameter of $d = 65.8\,\mathrm{cm}$ *upon whic*
t is raining. Water droplets collide inelastically with the wheel's surface, but subsequently the
an detach from it. What is the minimum angular rotation speed for water droplets to depar
rom the entire upper half of the wheel's perimeter?

Jindra rode his bike through the puddles

Problem 10 ... stealing with a spring

4 point

We attach a spring with a stiffness $k = 5.2\,\mathrm{N{\cdot}m^{-1}}$ *and a relaxed length* $l_0 = 15\,\mathrm{cm}$ *to a bod*
with mass $m = 120\,\mathrm{g}$. *We then start pulling at its other end at a constant speed* $v = 65\,\mathrm{cm{\cdot}s^{-1}}$
To what maximum length does the spring stretch? The motion takes place on a smoot
horizontal plane. *Jarda would like to become a pickpocke*

Problem 11 ... radiocarbon dating

5 point

Zuzka went to Abusir for archaeological excavations. In the tomb of an ancient Egyptia dignitary, she found a sample of wood, which she took to analyze on a mass spectrometer. Th ratio of carbon isotopes in the sample was measured as $p_{^{14}\mathrm{C}/^{12}\mathrm{C}} = 9.22 \cdot 10^{-13}$. *In what year wa the dignitary buried? Using the Gregorian calendar, write the years BCE with a minus sign The half-life of* ^{14}C *is* $T = 5\,730\,\mathrm{yr}$, *and the ratio of isotopes in the atmosphere is historicall constant* $p_0 = 1.25 \cdot 10^{-12}$. *Assume that the tree was cut down shortly before the dignitary burial. Jindra came up with the origin of the problem only after Terka pointed it out to him*

Problem 12 ... curious cyclist

4 point

Kuba bought a new bicycle and wanted to find out the size of the rolling resistance coefficient of his wheels. He noticed that the bike goes downhill on its own if the plane of the slop makes an angle with the horizontal greater than $\alpha = 0.5\,^\circ$. The diameter of the bicycle wheel is $d = 67\,\mathrm{cm}$. Can you help Kuba?

Kuba likes cycling

Problem 13 ... carrying a box

5 point

As Lego carried the box, he was pondering about force he was applying to it. The box ha a mass $m = 7.5\,\mathrm{kg}$ and is shaped like a rectangular cuboid. Lego holds it by pushing on tw opposite (vertical) sides, and the coefficient of friction between the sides and Lego's hand is $f = 0.45$. What is the magnitude of the force exerted by one of Lego's hands on the box?

Lego was carrying a lot of things at the cam

Problem 14 ... divider No.1

4 point

Petr needed a 3 V *voltage source, but he only had a* 12 V *source and resistors. So he decide*
o build a voltage divider from the schematic 4. What value of R *did Peter have to choose t*
et 3 V *at* V_+*?* *I wanted to reminisce about electrical engineering*

Problem 15 ... steel sphere floats

4 point

Karel found a steel sheet with a thickness of $\Delta r = 0.84\,\mathrm{mm}$ *and was thinking about what to d with it. Since Karel likes to experiment, he made a hollow sphere of such a radius that whe placed in the water, one half was above the surface and one below. Consider that the thicknes of the sphere is just* Δr *and that the density of the steel is* $\rho' = 7\,840\,\mathrm{kg{\cdot}m^{-3}}$*. Find the oute radius of this sphere.*

Karel was thinking on a boa

Problem 16 ... modified solar system

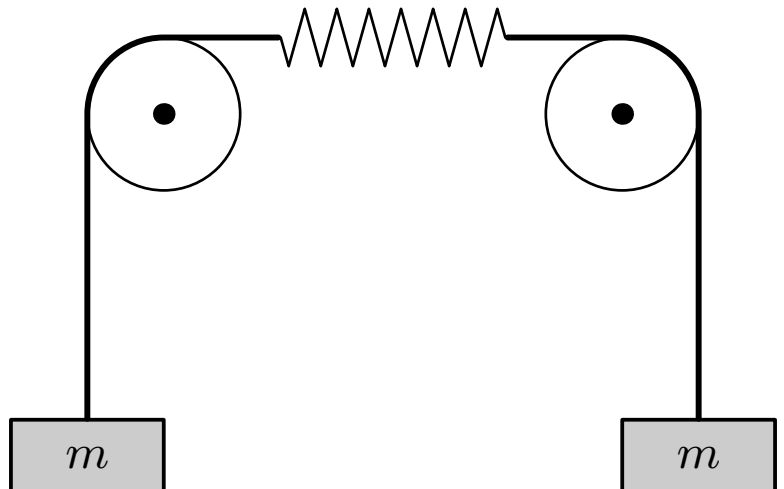
5 point

Suppose the Sun had an effective temperature $T_2 = 8\,000\,\mathrm{K}$. By how many percent would th
period of Jupiter's orbit have to be extended for the same amount of power to fall on it as i
does now? *Danka was trying hard to come up with an interesting problem*

Problem 17 ... pulleys with elastic rope

4 point

Lego likes tasks with pulleys. But this time he wanted to come up with a problem that would model the fact that the rope has some elasticity. He decided to do this by splitting a perfectly stiff, weightless rope in the middle and then joining it with a weightless spring with stiffness $k = 78\,\mathrm{N\cdot m^{-1}}$*. He then placed the modified rope on two fixed pulleys placed at the same height and hung a weight with mass* $m = 9.0\,\mathrm{kg}$ *on each end. He then held both the weights with just enough force so that the tension in the rope was* $T = 12\,\mathrm{N}$*. What is the acceleration with which the weights will move if they are released at the same time?*

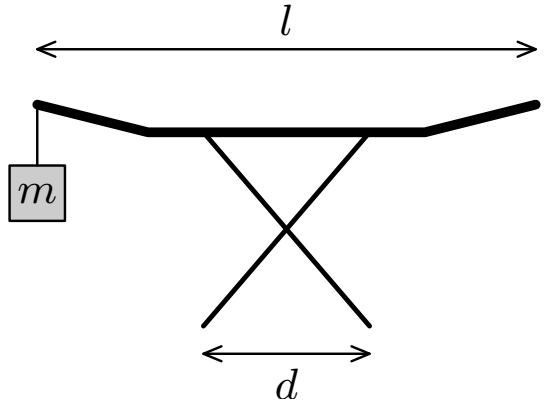


The problem came up to Lego's mind, when he had a lecture on pulleys at camp

Problem 18 ... falling clothes horse

4 point

Matěj has a clothes horse with a width of $l = 180\,\mathrm{cm}$ *on which he spreads his clothes evenly after washing. On the left edge of the dryer, he places a weight of* $m = 3\,\mathrm{kg}$*. How long after we hang up the clothes will the dryer tip over if its leg span is* $d = 60\,\mathrm{cm}$*? The mass of the wet clothes is* $M_0 = 6\,\mathrm{kg}$*; this will decrease to* $M_1 = 2\,\mathrm{kg}$ *when the clothes are completely dry. The weight of the dryer itself is* $2\,\mathrm{kg}$*. For simplicity, consider that the water evaporates at a constant rate and the clothes dry up in one day.*



Matěj cannot dry his clothes

Problem 19 ... acoustic speedometer

5 point

When the car is stationary, 80 *raindrops per second hit its windshield. What velocity is i raveling at if the current frequency of impacts is* $230\,\mathrm{s}^{-1}$*? The windshield has an area* S *and i nclined at an angle of* $33\,^\circ$ *with respect to the ground. Raindrops fall vertically to the groun t a velocity* $4.5\,\mathrm{m{\cdot}s}^{-1}$*.*

Jarda was afraid of a speeding fine

Problem 20 ... the Synchrotron in Grenoble

5 point

At the ESRF synchrotron in Grenoble, electrons with an energy of 6.03 GeV *orbit along th track with a circumference of* 844.4 m, *generating a current of* 35.8 mA. *How many electron are there in the entire synchrotron at one moment?*

Jarda participated in a diffraction experiment

Problem 21 ... maximal pétanque

6 point

Vojta played pétanque, but he was not good at it. Therefore, he got angry and threw the ba
he held in his hand as far away as he could. At what angle must it be thrown with velocity v =
= 11 m·s^{-1} to go as far as possible if the ball continues to roll after impact? When the ball hit
he ground, only the vertical component of the velocity is absorbed, the ball does not slip, an
he rolling resistance coefficient is $c = 0.17$. Assume that the ball falls on a nearby horizonta
plateau that is at the same height as was the ball when it left Vojta's hand.

Vojta misunderstood the rules of pétanque

Problem 22 ... taxidermy of a cylinder

5 point

Let's have a homogeneous cylinder. Around its axis, we cut a smaller cylinder out of it. Th
hollow and the smaller cylinder are then released down the inclined plane. What is the radius o
the smaller cylinder if it starts with 20 % more acceleration than the rest of the larger cylinder
Provide the answer in multiples of the original radius.

Jarda wanted to state a problem without any number. It didn't work ou

Problem 23 ... rats on rats

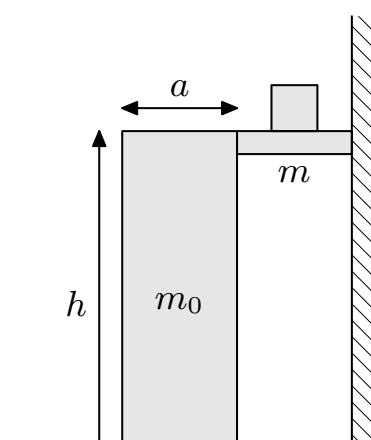
4 point

magine a heap of rats, each of which has mass m. We arrange the rats in a 2D pyramid simila
o Pascal's triangle. There will be one rat at the top, two rats below it, three in the next row
nd so on. We have a lot of them, infinitely many. Each rat distributes its weight and the weigh
t carries to the rats below it. What is the total weight the legs of the rat on the far left mus
arry? Express the result as a multiple of m. If it would be infinite, enter 0. Consider the rat
n a homogeneous gravitational field. *Karel was thinkig about Jára (da) Cimrma*

Problem 24 ... postmodern art

5 point

Lego wanted to create a sculpture. He therefore took a rectangular cuboid with mass $m_0 = 18.5\,\mathrm{kg}$, a square base with side length $a = 60.0\,\mathrm{cm}$ and height $h = 185\,\mathrm{cm}$, which he placed next to a solid wall (the sides of the cuboid are parallel to it). He then inserted a slab between the block (cuboid) and the wall so that it was held horizontally at height h only by friction. The coefficients of friction between the slab and the block, and the slab and the wall are $f = 0.42$. The coefficient of friction between the cuboid and the ground is effectively infinite. Lego, nevertheless, wants to put a weight on the middle of the slab. What is the largest sum of masses of the weights and the slab that the block can hold? The slab and the cuboid are homogeneous



Lego's problems are evolving

Problem 25 ... ice in a cube

5 point

We enclose ice with a mass of 15 g *and temperature* 0 °C *under normal conditions in a hermet*
cally sealed cubic container with a volume 0.10 l. *Afterwards, we start heating it. What wi*
be the pressure inside the cube when the temperature there reaches 120 °C*?*

Jarda wanted to trick the participants, but he tricked himself instead

Problem 26 ... When will I be there? 7 point

Jarda rides in an elevator that ascends at a constant speed. Since he is getting impatient, h
s throwing his keys into the air. They always fly to the height of 62 cm*. However, betwee*
ne ejection and the subsequent catching of the keys, the elevator begins to brake steadily, s
hat the keys fly to a height of 72 cm *and spend* 0.15 s *more time in the air. Determine th*
cceleration with which the elevator slowed down.

Jarda is in the elevator and already looking forward to bed

Problem 27 ... insidious horn

5 point

At the FYKOS camp, participants measured a car's speed using the frequency shift of th horn. However, they encountered a problem – the horn's pitch changed between individua repetitions. Lego, therefore, came up with the following modification of the experiment: w measured the horn's frequency when the car was approaching us as $f_1 = 437\,\mathrm{Hz}$*. Right afte that, the frequency* $f_2 = 415\,\mathrm{Hz}$ *was measured when the car passed us closely. Assuming tha neither the car's speed nor the frequency emitted by the horn has changed, what speed was th car traveling at? Lego really came up with the idea at the camp during the presentations*

Problem 28 ... incorrect voltage

5 point

A testing water electrolyzer produces 1.43 g *of hydrogen per hour. The device is powered by direct current from the source connected by cables, each having a resistance of* 3.1 mΩ. *Although the source exhibits an output voltage of* 1.95 V, *this value is not directly associated with electrolysis. What voltage would be measured if we connected a voltmeter directly to the electrolyzer?*

Jarda produces hydrogen

Problem 29 ... banded archerfish

6 point

Banded archerfish is a fish species that has found an original way to hunt for food. It approache the surface and spits out a stream of water to knock down unsuspecting insects nearby. Th insect falls into the water and has little time to escape. If the archerfish sees the insect sittin at an angle 35° *relative to surface normal, how far must the insect sit from it to be knocke down? Consider that the fish can knock down insects at a maximum height of* 3.0 m *above th surface.*

The fish splashed Jard

Problem 30 ... digging up

6 point

nside a hollow planet, a special life form has evolved. The inhabitants of this vacuum bubbl
vith a radius of $r = 1\,000\,\mathrm{km}$ *decided to dig their way up to the planet's surface. Thei*
cientists measured the density of the rock in several places and found that it decreases linearl
vith distance from the center of the planet. At the surface of their bubble, they measured
lensity of $9\,000\,\mathrm{kg{\cdot}m^{-3}}$ *and* $100\,\mathrm{km}$ *further from the centre they measured* $8\,800\,\mathrm{kg{\cdot}m^{-3}}$*. Thi*
gave them an estimate that their planet was unlikely to have a radius larger than $5\,000\,\mathrm{km}$*.*

While digging the tunnel to the planet's surface, they decided to take a lunch break at
listance $R = 3\,000\,\mathrm{km}$ *from the center of the planet. But they encountered a strange difficult*

gravity. There's zero gravitational acceleration inside their bubble, so they were surprised t
ind something pulling them back down. Calculate the gravitational acceleration at the poin
f their lunch break. Assume that all material at distance x from the center of the planet ha
he same density. *Kuba was reading The Wandering Earth*

Problem 31 ... echooooooooooo

6 point

n the middle of a long tunnel with radius $R = 15\,\mathrm{m}$ *stands a point source of sound which emit short beep. At a distance* $D = 210\,\mathrm{m}$*, also in the middle of the tunnel, we hear its intensit evel as* $60\,\mathrm{dB}$*. How long after we initially hear the beep do we stop hearing its echo if eac ime it reflects off the wall its intensity decreases by* $60\,\%$*? The lowest sound intensity that ca till be heard in the tunnel is* $22\,\mathrm{dB}$*.*

In his younger days, Jarda often visited caves of the Moravian Karst

Problem 32 ... maximal activity I

6 point

Jindra has $N_0 = 10^7$ atoms of the isotope ^{211}Bi. This isotope, with a half-life of $T_{\mathrm{Bi}} = 2.14\,\mathrm{min}$ transforms into the isotope ^{207}Tl, which then, with a half-life of $T_{\mathrm{Tl}} = 4.77\,\mathrm{min}$, transforms int the stable isotope ^{207}Pb. What is the maximal activity that the system can reach?

Jindra does not disdain any activity

Problem 33 ... immersing

5 point

We have a negligibly thin hollow sphere and a hollow cube, both with volume $V = 5\,\mathrm{l}$ and fu
of air. What is the ratio of the work required to submerge them under water? The base of th
ube is parallel to the surface (we don't rotate it during motion), both bodies are starting at th
urface, and we want to get them to a state where they are completely submerged and touchin
he surface. As a result, sumbit a number greater than 1. *Matěj was in the bathtub*

Problem 34 ... frog on a water lily

6 point

Jarda spends so much time in the garden that he enjoys getting to know its inhabitants However, a frog sitting on a water lily in the garden pond always gets scared and jumps away f the water lily leaf has a mass of 43 g *and the mass of the frog is* 150 g, *how far can it jump t jumps up to a distance of* 2.1 m *on solid ground? Assume that the water lily does not ben nd moves only in a horizontal direction and that it moves freely.*

Jarda is woken up by frog croaking at home

Problem 35 ... relativistic star

6 point

What would be the radius of a star with the same mass as the Sun, but its redshift would be s
arge that the wavelength of the radiation coming from its surface would double for an observe
t infinity? For simplicity, consider a non-rotating spherically symmetric star.

Karel was thinking about neutron stars and black holes

Problem 36 ... billiard

6 point

The FYKOS-bird is at the pool table, which is 3.5 by 7 feet and has a hole for balls in eac corner. There are white and green balls on the table, which are otherwise the same, fittin just right into the holes and bouncing off the walls and each other with perfect elasticity. Th FYKOS-bird originally placed the green one on the longer axis of symmetry of the table. H nudged the white towards the green. The green crashed into one of the corner holes. Howeve After several impacts from the longer edges of the table, the white ball ended up in the sam hole. How many positions on the table are there to place the green ball for this situation t play out? Ignore the rotation of the balls.

Jarda mostly hits the hole with white

Problem 37 ... impedance spectroscopy

7 point

In electrochemical experiments, we often encounter a method of sending an AC signal to th device, we then change its frequency and monitor the evolution of the impedance. Conside An alternative circuit, which consists of a parallel connection of a capacitor with capacitance C and a resistor with R_2, which are connected in series with R_1. Determine the largest possibl phase shift by which the voltage is delayed behind the current if $4R_1 = R_2$.

Jarda examined data from his bachelor's thesis on electrolyzers

Problem 38 ... small hole in the water balloon

7 point

We have a hollow sphere with thin walls of radius $R = 10.5\,\mathrm{cm}$. This sphere lies on (or i ttached to at its lowest point) a horizontal plane; it is completely filled with water and ha hole at its highest point. Where do we need to drill another hole so that the water will spra ut as far as possible from the point of contact of the sphere with the plane? We make th ole infinitely small. Enter the result as the angle formed by the line joining this hole and th enter of the sphere with the vertical direction (i.e., a vector perpendicular to the plane an ointing away from it forms an angle of 0). Lego thought, that a nice problem on hydro. .

Problem 39 ... changing of our star

6 point

Consider a star similar to our sun that emitted most of its energy at a wavelength of 504.7 nm *and had a radius corresponding to* $0.990R_\odot$*, where* $R_\odot$ *is the radius of today's Sun. Ove time, its composition has changed, increasing its radius to* $R_\odot$*, while shifting the wavelengt*

of maximum emission by 3.7 nm *towards the ultraviolet part of the spectrum. How much ha*
he luminosity of this star increased during this time?

Karel wondered about the history and future of the Sun

Problem 40 ... top of the class

6 point

During one of the boring lessons at school, the students invented their own fun – they threw a small heavy object perpendicularly upwards so that it would be as close to the ceiling as possible, but at the same time, would not touch it. The ceiling is at a height $H = 2.7\,\mathrm{m}$ from the point of the throw. However, when they did this fun in physics class, the teacher made them measure the initial velocity and its standard deviation as $(7.0 \pm 0.5)\ \mathrm{m\cdot s^{-1}}$. How likely are the students to hit the ceiling if the distribution of initial velocities is Gaussian?

Jarda was listening to the Vašek's stories

Problem 41 ... pressure difference

6 point

Jindra has a solid cylindrical tube with a $H = 50.0\,\mathrm{cm}$ *length and a radius of* $r = 4.00\,\mathrm{cm}$ *One end of the tube is airtight sealed with a lid. Then, he held the tube by the lid an began to submerge it perpendicularly beneath the sea level with the open end. What will b the maximum difference between the pressures acting on the top and bottom lids of the tub during submersion? Jindra submerged the tube to a depth allowed by the length of his arm* $= 65.0\,\mathrm{cm}$. *The density of water is* $\rho = 1\,024\,\mathrm{kg{\cdot}m^{-3}}$. *The air at sea level is under norma conditions. Both sea water and air have the same temperature.*

Jindra didn't know how else to entertain himself on a holiday by the sea

Problem 42 ... fireworks

6 point

After the competition, the FYKOS organizers planned a fireworks display. They will launc their firework at $45\,\mathrm{m\cdot s^{-1}}$ *perpendicularly upwards, which will disintegrate into many sma pieces in* $3.3\,\mathrm{s}$. *These fly off in all directions from the point of disintegration at* $15\,\mathrm{m\cdot s^{-}}$ *relative to the reference frame of the original firework and glow for* $5.5\,\mathrm{s}$. *Determine the volum of space into which the fragments have managed to spread when they are extinguished.*

Jarda likes to toast on fireworks

Problem 43 ... electron collision

6 point

n one accelerator, colliding electrons are flying towards each other, each with an energy o
04.5 GeV. In a second accelerator, electrons with an energy of 209 GeV are flying toward
target made of stationary electrons. How many times more energy is available for the creatio
f matter in the first accelerator compared to the second?

Jindra felt like colliding his electrons

Problem 44 ... we want to breathe on Everest

7 point

What would be the air pressure at sea level if the Earth had an atmosphere with the same tem
perature lapse rate as it does today (i.e., a linear decrease of temperature by $0.65\,^{\circ}\mathrm{C}$ *per* $100\,\mathrm{m}$
altitude), but at an altitude $H = 8\,850\,\mathrm{m}$ *above sea level (on Mount Everest), the pressure p*
would be the same as it is today at sea level? Consider the sea level temperature $T_0 = 15\,^{\circ}\mathrm{C}$.

Karel was thinking about the atmosphere again

Problem 45 ... a lake on a mirror reloaded

7 point

On a table, Jindra laid a hollow spherical mirror with a radius of curvature of $r = 2.00\,\mathrm{m}$ *and a diameter* $D = 12.0\,\mathrm{cm}$. *The optical axis of the mirror points upwards. On the mirror he poured water with a refractive index* $n = 1.33$ *so that the water formed a surface with a diameter* $d = 6.00\,\mathrm{cm}$. *Jindra illuminated the whole surface of the mirror with rays of light parallel to the optical axis. At what height above the table would an image with the smallest outer diameter possible be created on a hypothetical screen?*

Jindra misses summer swimming

Problem 46 ... coin funnel

7 point

Consider a heavy hemispherical bowl of 19 cm *in diameter and take a point mass that can mov*
n the bowl completely frictionless. We release it from the bowl's edge with velocity $0.8\,\mathrm{m{\cdot}s^{-}}$
n the horizontal direction. What is the minimum height above the bottom of the bowl that th
oint mass can reach as it travels through the bowl? *Jarda's wallet spilled out on the stree*

Problem 47 ... a point jumps over a cuboid

7 point

Once upon a time in the land of geometric shapes, a point mass was sitting still on a horizonta
plane when it suddenly spotted a cuboid $h = 20.0\,\mathrm{cm}$ *tall and* $l = 280\,\mathrm{cm}$ *long approaching i*
with a speed $v_k = 6.80\,\mathrm{m{\cdot}s^{-1}}$. *Our poor point mass realizes the time to jump away ran out*
so it just jumps over it. What is the minimum speed that the point mass needs to jump i
order to be able to jump over the block? The point mass does not only need to jump verticall
upwards. *This idea has been in Lego´s ideapad for such a long tim*

Problem 48 ... oscillating lens

8 point

Let's place an isotropic light source on the optical axis of a lens with a focal length of $f = 8.5\,\mathrm{cm}$ *at a distance* $A = 11\,\mathrm{cm}$ *from its center. Then, we attach a spring to the lens, allowing it to perform torsional oscillations. The axis of oscillation is perpendicular to the optical axis of the lens. The moment of inertia of the lens with respect to this axis is* $J = 63\,\mathrm{kg{\cdot}mm^2}$*, and the torque acting on the lens is proportional to the angle of rotation from the equilibrium position as* $M = -c\varphi$*, where* $c = 3.7\,\mathrm{mJ}$*. What is the maximum speed at which the image of a point light moves if the lens is deflected* 5° *from its equilibrium position and let go?*

Jarda wanted to combine optics and oscillation

Problem 49 . . . epic fail

8 point

Radek and Radka are merrily enjoying the ride on the carousel, with Radka sitting a quarte ircle in front of Radek (in the direction of rotation). In a moment of mischief, Radka throws otten tomato directly at Radek, but after less than half of the carousel's rotation period, th omato appears back in Radka's face. Determine the magnitude of the velocity (as the ratio o > 0 to the circumferential velocity of the carousel) with which Radka threw the tomato. Thi rotesque took place in weightlessness. Radka didn't want to hit. . . Didn't want to hit herself

Problem 50 ... intermittent voltage

7 point

n the circuit, a capacitor with capacitance $C = 47\,\mathrm{nF}$ *and a resistor with resistance* $R = 220\,\mathrm{k}\Omega$ *re connected in series to an intermittent voltage source with period* $2T = 20\,\mathrm{ms}$. *The voltag waveform over one period is*

$$V(t) = \begin{cases} 0\,\mathrm{V} & -T < t < 0; \\ 9.0\,\mathrm{V} & 0 \le t < T. \end{cases}$$

Determine the difference between the highest and lowest voltage on the capacitor at steadj tate.

Jindra was playing with a switch in a DC circuit

Problem 51 ... Van Allenovi is just bright

8 point

n the equatorial plane, at a distance of two Earth radii from the center of Earth, there i proton with energy $E = 1\,\mathrm{keV}$, which is pointing at an angle $\alpha = 45^\circ$ from the force lin owards the North Pole. As it approaches Earth, at one moment it is reflected by a magneti mirror and returns towards the South Pole. Determine the (magnetic) latitude at which thi appens. Treat the Earth's magnetic field as a dipole.

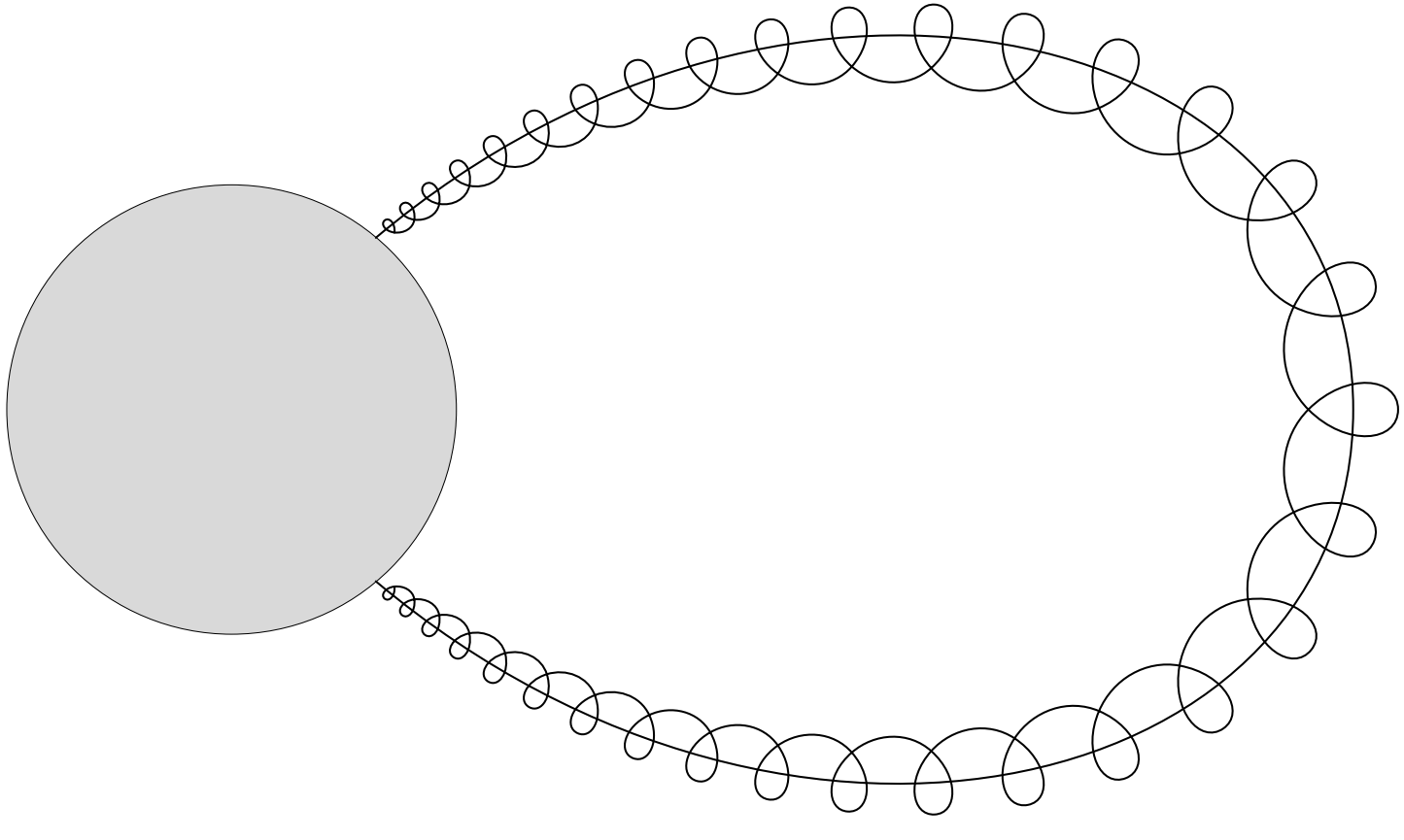


Figure 8: Representation of proton behavior.

Kačka got the result, but wanted to check it

Problem 52 ... flying mud

8 point

A tire with an outer diameter $d = 63.2\,\mathrm{cm}$ *is rolling on a flat surface at a constant speed* $v =$ $= 15.0\,\mathrm{m{\cdot}s^{-1}}$. *Suddenly, a piece of mud gets ejected from its rotating circumference. The so lies through the air and eventually lands on the ground. Subsequently, the tire passes over th fallen piece of mud with the same part from which it was ejected. How long was the soil on th ground before the tire ran over it? Disregard air resistance during the flight of the mud. Th mud did not fall farther than* $4.00\,\mathrm{m}$ *along the horizontal axis from the release point.*

While driving, a fallen tractor wheel whizzed past Jindra

Problem 53 ... maximal activity II

9 point

Jindra has $N_0 = 10^9$ atoms of the radioactive isotope ^{212}Bi. It decays to the isotope ^{212}Po wit probability $P_\beta = 64.06\%$ by beta decay and to the isotope ^{208}Tl with probability $P_\alpha = 35.94\%$ by alpha decay. The half-life of bismuth is $T_{\mathrm{Bi}} = 60.6\,\mathrm{min}$. The polonium isotope decays furthe by alpha decay with a half-life of $T_{\mathrm{Po}} = 299\,\mathrm{ns}$ to the stable isotope ^{208}Pb. The thallium isotop decays by beta decay with a half-life of $T_{\mathrm{Tl}} = 3.05\,\mathrm{min}$ also to lead ^{208}Pb.

How long does it take Jindra to measure the maximum activity in the system?

Jindra measured the half-life of ^{212}Po

Problem 54 ... Mišo is shooting

9 point

Mišo likes to shoot from a laser. He needs to reduce its energy with reflective gray (neutra
lensity, ND) filters. He would like to achieve a beam with an energy 37 J*. Mišo has 5 filte*
olders and 7 different filters, namely 2-stop ND, 3-stop ND, 5-stop ND, 7-stop ND, 11-sto
ND, 13-stop ND, and 17-stop ND. Assume that all energy striking the filter is either reflecte
r transmitted. The laser has an energy of 77 377 J*. With what accuracy can he achiev*
he desired 37 J*? Give the result in* mJ*.* *Mišo was calculating the filtering at the PAL*

Problem 55 ... Red Wednesday

9 point

Lego is situated aboard a space station in deep space. To avoid going crazy, he keeps watchin what is the day today on Earth. He knows that last Sunday he launched a rocket (at rest) wit rest mass $m_0 = 31.0\,\mathrm{kg}$ propulsed by an ion thruster which does not reduce the rest mass however, it exerts a constant force F. The rocket glows with a golden light $\lambda_0 = 600\,\mathrm{nm}$ i such a way that when Lego points his telescope at it today at midday (specifically $t_\mathrm{s} = 72.0$ ince the launch), he sees a red light $\lambda_r = 670\,\mathrm{nm}$. How big is the force F?

Lego spotted a pattern in the date of Physics Brawl Online...

Problem 56 ... pumping water using water

8 point

n the 19th century, people began to make extensive use of steam engines and to think abou heir efficiency. One of the problems they could solve back then was pumping water out of th mine. However, we are not interested in the specific design of a process/engine that pumps wate rom a depth of $h = 50\,\mathrm{m}$ *to the surface. Consider that we have stumbled upon geothermall eated ideal water (it is incompressible, has a constant density* ρ *and specific heat capacity c with volume* $V_1 = 200\,\mathrm{m}^3$ *and temperature* $T_1 = 90\,^\circ\mathrm{C}$*. Nearby is also a lake (thermal bath with constant temperature* $T_\mathrm{J} = 10\,^\circ\mathrm{C}$*. What is the maximum amount of water we can ideall ump to the surface? For the whole amount of water, consider a constant elevation* h *an homogeneous gravitational field.*

Marek made an excursion to the 19th century

Problem 57 ... a beer problem

9 point

Drinking beer is not as simple as it seems. You must pick up your half-pint and tilt it so muc
that it starts pouring into your mouth. For this, however, work must be done. Consider th
half-liter weight of $m = 360\,\mathrm{g}$, *its cylindrical shape of radius* $r = 3.5\,\mathrm{cm}$ *and height* $h = 15\,\mathrm{cm}$
and density of beer $\rho = 1\,030\,\mathrm{kg{\cdot}m^{-3}}$. *Determine the work done if you lift a half-liter fille*
with $350\,\mathrm{ml}$ *of beer to drink. Your mouth is* $40\,\mathrm{cm}$ *above the table.*

Jarda was thinking about how much he has to pay Viktor again. .

Problem K.1 ... the pendulum swings back... 3 point

n certain buildings, one might encounter Foucault's pendulum, which has a very long suspen ion. Jarda once encountered one, wanting to estimate the height of the ceiling from whic he pendulum was suspended. He measured the period of oscillation as 14.2 s*, and when th endulum was in its lowest position, it was* 70 cm *above the floor. Determine the height of th eiling.*

Jarda has been in the Panthéon in Paris

Problem K.2 ... ...and forth...

3 point

To be able to use the approximation of a mathematical pendulum, we need a point of mas
on an massless suspension. Let's consider a string made of steel with a diameter of 1.4 mm *o*
which a weight is suspended. For good accuracy, we require it to be 80 *times more massive tha*
the string on which it is suspended. If the tensile strength of the used steel is $520\,\mathrm{N{\cdot}mm^{-2}}$
what can be the maximum length of the suspension at rest? The density of the used steel i
$7\,900\,\mathrm{kg{\cdot}m^{-3}}$. *Such a thin and long rope, and yet it would still hold Jarda*

Problem K.3 ... ...and back...

4 point

The horizontal displacement of a pendulum has lowered from 1.0 m *to* 0.9 m *in ten minutes Determine the average work the resistance forces do during a single pendulum period. Again ssume a mathematical pendulum with a* 47 kg *weight and a period of oscillation of* 14.2 s.

Jirka thought the statement of this problem was confusing

Problem K.4 ... ...and forth, over and over again

5 point

The estimate did not seem accurate enough to Jarda, so he decided not to consider the pendulu
s a simple mathematical pendulum but went on to include the moments of inertia of its part
n his calculations. He found that the suspension consisted of 50.0 m *long string made fro*
teel with a diameter of 1.40 mm *and that at the end of the suspension, there was a ball o*
adius 10.0 cm *with mass* 17.0 kg*. What is the ratio of the period of such a pendulum to th*
eriod of a mathematical pendulum of length 50.1 m*? The density of steel is* $7.90\,\mathrm{g\cdot cm^{-3}}$*.*

Jarda needed a physical pendulum problem also for this year's Hurry-up

Problem T.1 ... inflating a balloon

4 point

When we are inflating a balloon with a large enough radius, we can accurately assume th balloon's elastic potential energy to be proportional to its surface area. Jirka found such balloon with a radius of 5 cm *and measured the air pressure inside it to be equal to* 107 kPa *However, the balloon was not very large, so he decided to inflate it to a radius of* 15 cm*. How many moles of air did he need to blow into the balloon? Assume a spherical balloon surrounde by room air with a temperature* $T = 20\,^{\circ}\mathrm{C}$*.* *Optics tutorial was too boring for Jirka*

Problem T.2 ... we are cooling down the balloon

5 point

Jirka was already satisfied with the size of his balloon and took it for a walk. He left the heate oom at 20 °C *wearing only a T-shirt and found out that he was really cold because the outsid emperature was only* 3 °C*. What is the new radius of the balloon after it shrank during th valk if it initially had a radius of* $r_1 = 15\,\mathrm{cm}$*? Do not forget that in the previous problem ou derived a relationship between the overpressure in the balloon and its radius (assuming th balloon has a spherical shape)*

$$\Delta p = \frac{A}{r}\,,$$

where $A = 300\,\mathrm{Pa{\cdot}m}$.

Jirka attends secret night meetings

Problem T.3 ... our balloon has flown away

6 point

Viktor had gifted Jirka a little present for his drive during proofreading – a helium balloo with radius $r_0 = 11\,\mathrm{cm}$. *Jirka weighted down the balloon in a way that the total weight o the material was* $M = 5.2\,\mathrm{g}$ *and hoped that it was enough for the balloon not to fly away Unfortunately, he was mistaken, and the balloon started to rise. If we consider the sam dependency between the radius and the pressure inside the balloon as in the problems before .e.*

$$\Delta p = \frac{A}{r}\,,$$

where $A = 300\,\mathrm{Pa{\cdot}m}$, *determine to what height the balloon will rise in an isothermic atmo sphere with a temperature* $T = 20\,^\circ\mathrm{C}$, *provided that it won't pop. The molar mass of heliu s* $4.003\,\mathrm{g{\cdot}mol^{-1}}$.

Hint: *In an isothermic atmosphere both the pressure and the density decrease exponentially*

Actually, Jarda invited Jirka to have a bee

Problem T.4 ... connecting balloons

6 point

Because Jirka's helium balloon had flown away, he sadly had to ask Viktor for another one He got two, but he had to inflate them himself. Jirka inflated one to a radius of $r_i = 15.0\,\mathrm{cm}$ *hen took a short, narrow tube and connected it to the other so that no air escaped. To hi reat surprise, once the balance was established, one balloon had a larger radius than the other What was the ratio of the radius of the larger balloon to the radius of the smaller one? Now uppose that the pressure difference between the balloon and the surrounding area depends o ts radius as*

$$\Delta p = \frac{A}{r}\left[1 - \left(\frac{r_0}{r}\right)^6\right],$$

where $A = 300\,\mathrm{Pa{\cdot}m}$ *and* $r_0 = 3.00\,\mathrm{cm}$*. The experiment was conducted at a* $20\,^\circ\mathrm{C}$*. Assume th econd balloon has radius* r_0 *before attaching it to the tube.*

Jarda saw an interestig experiment at the lecture

Problem M.1 ... tram

3 point

What is the maximum angle at which a tram can travel downhill and still be able to stop? Th
oefficient of shear friction of the wheels and rails is $f = 0.15$.

David rode to the lecture suspiciously downhil

Problem M.2 ... the tram reloaded

3 point

The tram travels at $v = 27\,\mathrm{km{\cdot}h^{-1}}$ and its maximum deceleration is $a = 2.1\,\mathrm{m{\cdot}s^{-2}}$. What i
he minimum distance required for the tram to come to a complete stop upon Davis's signalin
t the stop? The reaction time of the driver to the David's wave is $t' = 0.3\,\mathrm{s}$.

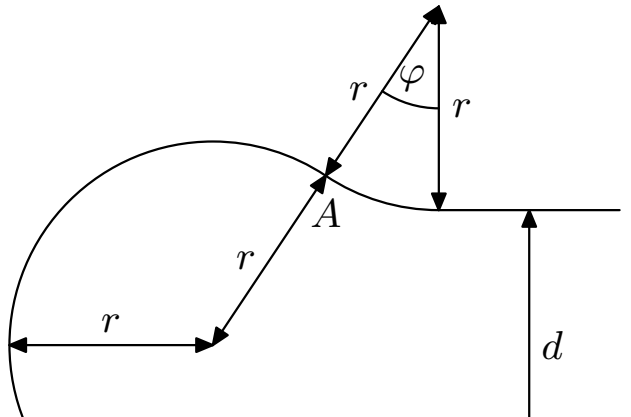
David studied the T3 tram manual for far too lon

Problem M.3 ... parallel tracks

4 point

Consider two parallel tram tracks at a distance of $d = 11\,\mathrm{m}$. *We want to build a system of arcs between them with a radius* r, *as shown in the right image. Find* r *such that a tram, capable of withstanding a centripetal acceleration of* $a = 0.85\,\mathrm{m{\cdot}s^{-2}}$ *in turn, passes through it in the shortest possible time. The velocity of the tram remains constant.*

Adam would like to ride the Snowpiercer.



Problem M.4 ... turboflies' troubles

5 point

Two trams at a distance of $s = 400\,\mathrm{m}$ *are moving against each other, first with a velocity o* $v_1 = 30\,\mathrm{km{\cdot}h^{-1}}$, *second with a velocity of* $v_2 = 35\,\mathrm{km{\cdot}h^{-1}}$. *From the first one, a fly take off with a velocity of* $v_3 = 80\,\mathrm{km{\cdot}h^{-1}}$ *and flies to the other tram, where it bounces off th windshield, flies back, and so on, as long as it manages to outfly the trams. The windshield are sticky, so with each bounce, its speed gets lowered to a* q*-factor of the velocity before th bounce, where* $q = 0.97$. *Assuming the bounce is instantaneous, what distance does the fl cover before the trams collide?*

Marek has seen a fly in a tram